Project Name	Ontology-Driven Operating System (ODOS)
Online team meeting	https://tu-berlin.zoom-x.de/j/66022239802?pwd=I409Z0ib9RwajWU4sIf7ySD1WmR0Xa.1
Production system (if any)	...
Test system (if any)	...
GitHub repository	https://github.com/amosproj/amos2026ws04-odos
GitHub feature board	https://github.com/orgs/amosproj/projects/112/views/2
GitHub imp-squared backlog	https://github.com/orgs/amosproj/projects/116
Position of T-shirt logo	Upper Torso Right Small
Link to logo for black T-shirt	https://github.com/amosproj/amos2026ws04-odos/blob/main/Deliverables/sprint-01/team-logo-black-shirt.svg
Link to logo for white T-shirt	https://github.com/amosproj/amos2026ws04-odos/blob/main/Deliverables/sprint-01/team-logo.svg
Additional materials	...
Team maling list	oss-amos-proj4@lists.fau.de

Last Name	First Name	GitHub User Name	Email Address
Haykır	Zeynep	zeynephaykir	z.haykir@campus.tu-berlin.de
Thomits	Nicole	NicoleT02	nicole.thomits@fau.de
Pervana	Sara	saramakishti	sara.pervana@campus.tu-berlin.de ; saramakishti
Akram	Wasim	WasimAkramShuvo	wasim.akram@fau.de
Barvencik	Moritz	moritzbarv	barvencik@campus.tu-berlin.de ; barvencik.moritz@web.de
Hirschpeck	Leon	leones18	leon.hirschpeck@fau.de
Li	Linfeng	WindyLeev	linfeng.li@campus.tu-berlin.de
Soliman	Yara	YaraSoli	y.soliman@campus.tu-berlin.de

		Product Owner					
#	Meeting Day	Review	Planning	Software Developer	Release Manager	Scrum Master	Comment
1	2026-10-14	--	Zeynep	Everyone else	N/A	Dirk	
2	2026-10-21	Zeynep	Nicole	Everyone else	Moritz	Sawera	
3	2026-10-28	Nicole	Zeynep	Everyone else	Yara	Sara	
4	2026-11-04	Zeynep	Nicole	Everyone else	Moritz	Sara	
5	2026-11-11	Nicole	Zeynep	Everyone else	Leon	Sara	
6	2026-11-18	Zeynep	Nicole	Everyone else	TBD	Sara	
7	2026-11-25	Nicole	Zeynep	Everyone else	TBD	Sara	Mid-term due
8	2026-12-02	Zeynep	Nicole	Everyone else	TBD	Sara	
9	2026-12-09	Nicole	Zeynep	Everyone else	TBD	Sara	
10	2026-12-16	Zeynep	Nicole	Everyone else	TBD	Sara	
11	2027-01-13	Nicole	Zeynep	Everyone else	TBD	Sara	
12	2027-01-20	Zeynep	Nicole	Everyone else	TBD	Sara	
13	2027-01-27	Nicole	Zeynep	Everyone else	TBD	Sara	
14	2027-02-03	Zeynep	Nicole	Everyone else	TBD	Sara	Demo day!
15	2027-02-10	Nicole	Zeynep	Everyone else	TBD	Sara	Retrospective
Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here							

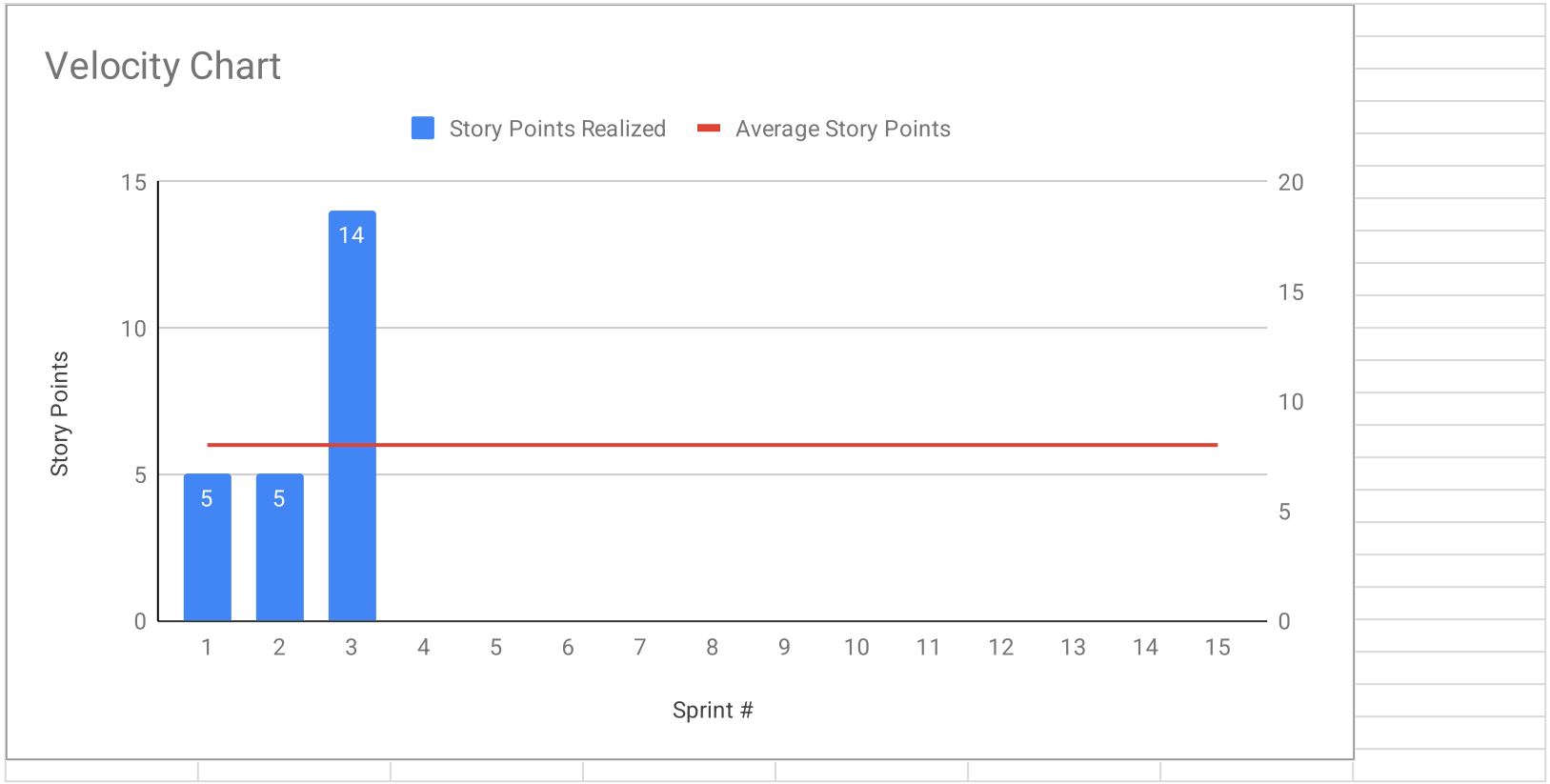
Goals	Ensure everyone learns the AMOS lessons
	Helping and learning from each other
	Ensure happiness with final result
	Have a good team experience
	Deliver an MVP and successfully complete the project
	Comply with relevant standards
Meeting norms	Be punctual
Working norms	Complete adopted sprint backlog items by the end of a sprint
	Decide based on consensus (everyone but one needs to agree)
Coordination norms	Communicate information in time e.g. for meetings you might miss
Communication norms	Be open and honest about what needs saying
Consideration norms	Be respectful and considerate about others
Cont. improvement norms	Regularly update the team on progress
Rewards	
Sanctions	(After one warning) talk to the authorities
Signatures	
Scrum Master	Sara Pervana
Product owner	Nicole Thomits
Product owner	Zeynep Haykır
Software developer	Leon Hirschpeck
Software developer	Moritz Barvencik
Software developer	Linfeng Li
Software developer	Wasim Akram
Software developer	Yara Soliman

Product Vision	Project Mission
Small and medium-sized industrial companies need software that fits their own products and processes, but they can’t afford to build it from scratch. ODOS gives them a digital twin of their product or plant that they can trust. Laws, standards and company rules are part of the twin, so only designs that follow them are accepted.	In this project we turn laws and standards into rules that a computer can check. We start with two documents, the EU Machinery Regulation and the German Musterbauordnung. We read their PDFs into a structured JSON, extract the rules from it, and check digital twins against those rules, so that only designs that follow the law pass. For Wisdom Bridge AI, we deliver a clear recommendation and a working prototype.

Term	Definition
D.	-
Digital Twin	a virtual representation of a physical object or process
E.	-
ERP	(sw.) Enterprise Resource Planning
M.	-
MES	(sw.) Manufacturing Execution System
O.	-
Ontology	a formal model for a specific domain, describing the terms (classes), properties and relationships which exist in said context
P.	-
PLM	Product Lifecycle Management

Sprint #	Sprint goal
1	Set up the team and its ways of working, and establish contact with the business partner.
2	Present researched options for representing normative documents to the business partner and agree on which ones to explore further.
3	In depth research on Akoma and SHACL, have a meeting with the business partner & receive feedback. Start working on the JSON schema
4	Pull ahead tasks from future sprints (focusing especially on DoDs), set up the foundation for the mid-project release by ensuring base functionality is stable
5	TBD
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	

Sprint #	Story Points Realized	Average Story Points
1	5	8
2	5	8
3	14	8
4		8
5		8
6		8
7		8
8		8
9		8
10		8
11		8
12		8
13		8
14		8
15		8
	PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB	


Velocity Chart
Story Points Realized
Average Story Points
Story Points
Sprint #
0
5
10
15
20
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
5
5
14

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
1			0	0	0	0
2			0	0	0	0
3			0	0	0	0
...				0		0
Features						
1						
2						
3						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

Sprint	Goal	Feature Name	Est. size	Est. remaining	Real size	Real remaining
Release						
Total			0	0		
Sprints						
1			0	0	0	0
2			0	0	0	0
3			0	0	0	0
...				0		0
Features						
1						
2						
3						
		PLEASE CREATE THE BURNDOWN CHART ON A NEW TAB USING THE DATA FROM THIS TAB				

#	Feature Definition of Done	Sprint Release Definition of Done	Project Release Definition of Done
1	All test cases for the feature pass	All planned and implemented feature branches were successfully merged into develop	All deliverables for past and the current sprints are up to date, complete and on the repo
2	The code follows Python coding conventions (see PEP 8)	All issues on the board are in their proper status and finalized issues have their real size assigned	Stale branches have been deleted
3	No commented (unused) code, open TODOs or debugging instructions/output	The software is running with no errors and everything is properly demonstrated during the review	The develop branch has been successfully merged into the main branch
4	A pull request has been submitted and reviewed by at least one other developer	-	The proper tags and versioning are set (CalVer is the simplest approach: Year.Month.Patch - e.g. "v26.11.2" or see Ubuntu release history)
5	Commit and pull request are linked to the corresponding GitHub issue	-	No sensitive data is on the repo (!.gitignore)
6	-	-	The formal review and demonstration took place in front of the POs and Stakeholders

Type	Link / reference

#	Context	Name	Version	License	Comment
1	Development / Runtime	Python	3.12	PSF-2.0	IN USE – Backend runtime
2	Document extraction	Docling	2.143.0	MIT	IN USE – PDF extraction
3	Data model	Pydantic	2.14.1	MIT	IN USE – DocumentModel and data validation
4	JSON validation	jsonschema	4.26.0	MIT	IN USE – Legal JSON 2.0 schema validation
5	File uploads	python-multipart	0.0.33	Apache-2.0	IN USE – PDF file uploads
6	PDF viewer	pdfjs-dist	6.4.299	Apache-2.0	IN USE – PDF viewer and highlighting
7	LLM Integration	Anthropic	2.3.0	MIT	PLANNED – LLM integration for rule extraction
8	Backend framework	Fastapi	0.162.0	MIT	IN USE – Backend API
9	Backend server	Uvicorn	0.54.0	BSD-3-Clause	IN USE – Running the FastAPI backend
10	Backend testing	Pytest	9.1.1	MIT	PLANNED - TESTING
11	Frontend UI	React	19.3.1	MIT	IN USE – Frontend UI
12	Frontend tooling	TypeScript	7.0.2	Apache-2.0	IN USE – Frontend type checking
13	Frontend tooling	Vite	8.3.5	MIT	IN USE – Frontend development and build
14	Frontend testing	Vitest	5.0.5	MIT	IN USE – Frontend testing
15	Database	MongoDB	9.0.4	SSPL-1.0	PLANNED – Primary database option
16	Database	PostgreSQL	18.6	PostgreSQL License	PLANNED – Alternative database option

Last Name	First Name	Value					
Haykır	Zeynep			#DIV/0!	#DIV/0!		
Thomits	Nicole						
Pervana	Sara						
Akram	Wasim						
Barvencik	Moritz			0	No size		
Hirschpeck	Leon			1	Trivial size		
Li	Linfeng			2	Small size		
Soliman	Yara			3	Medium size		
				5	Large size		
				8	Very large size		
				13	Too large (size)		

How to play planning poker

1. Everyone type their number into their value field, don't hit return yet
2. Someone, perhaps a product owner, count down 3.. 2.. 1..
3. Then, everyone hit return to submit their value

Efficient approach

1. Estimate the story points for the current ticket
2. Once ready, click and type your estimation into the cell under "Value" which corresponds to you and wait for the signal to hit ENTER
3. In case of a new round, the MC resets the fields, everyone clicks away and go to 1.